



Dr. Sovan Sankalp

Postdoctoral Researcher

AHRC, IIT Bhubaneswar, Odisha

Personal Profile

Date of Birth: 03.01.1991

Phone: 7008038833 / 8763211486

Email: sankalpsovan9@gmail.com

Address: Unit 9, Bhubaneswar, Odisha

Marital Status: Married

Languages: English, Hindi and Oriya

Education

- **Ph.D. in Civil Engineering** (2016-2023),
NIT Rourkela, Odisha, India. **CGPA-9.2**
Specialization: Water Resources Engg.
- **M.Tech in Civil Engineering** (2013-2015), NIT Rourkela, Odisha. **CGPA-8.62**
Specialization: Water Resources Engg.
- **B.Tech in Civil Engineering** (2008-2012)
SOA University, Odisha. **CGPA-8.30**
- **Intermediate** (2006-2008), CBSE,
Zinc Smelter Sr.Sec School, Visakhapatnam
- **Matriculation** (Class of 2006), ICSE
Blessed Sacrament High School, Puri

Technical Expertise

Arc GIS, ENVI, ERDAS IMAGINE, Google Earth Engine.

Python (Jupyter Notebook), R-Studio, R
HEC-RAS, HEC-HMS, CES, CCHE-2D

Research Interests

- Hydrological Analysis and Modelling
- Groundwater Hydrology
- Flood Analysis using MCDM approach
- Imperviousness and impact on Climate Change.
- Time series Forecasting using AI & ML
- Spatial and temporal Drought monitoring
- Land surface Temperature Monitoring

Academic Projects

Ph.D. Project

Title: Urbanization and its impact on hydrologic and environmental parameters of urban Indian catchments. **Supervisor:** Dr Sanat Nalini Sahoo, Assistant Professor

M.Tech Project

Title: Experimental and Numerical Investigation of Rough Bed Meandering Channel. **Supervisor:** Dr Kishanjit Kumar Khatua, Professor

B.Tech Project

Title: Utilization of Ferrochrome Slag as a Construction Material.

Supervisor: Dr Chittaranjan Panda, Assistant Professor

Publications

Science Citation Indexed Journals

- **Kisi, O., Heddam, S., Sankalp, S., Petroselli, A., Külls, C., & Zounemat-Kermani, M. (2026).** Cluster-based XGBoost framework for short-term rainfall–runoff prediction under uncertainty in the Sieber watershed, Germany. *Journal of Hydrology: Regional Studies. (Accepted) (IF:5.0).*
- **Dandapat, A. K., Panda, P. K., Sankalp, S., Kisi, O., Kraiem, H., Kucher, O. D., & Tariq, A. (2025).** “Ensemble deep learning framework for groundwater storage forecasting under hydrological variability”. *Acta Geophysica, 74(1). (IF:2.1).*
- **Dandapat, A. K., Panda, P. K., Sankalp, S., & Jothiman, M. (2025).** “Evaluating drought risks and meteorological hazards in the Mahanadi basin: Impact of precipitation deficits and climate change on the downstream regions of Hirakud Dam.” *Discover Applied Sciences. (ESCI).*
- **Cengiz, T. M., Kisi, O., Vishwakarma, D. K., Sankalp, S., Bassukas, D., & Külls, C. (2025).** “Spatial-temporal variation of precipitation in Niedersachsen, Germany, using statistical and graphical trend methods considering climatic drivers.” *Hydrological Sciences Journal (IF: 2.8).*
- **Sankalp, S., Sahoo, B. B., Gupta, S. K., Bhushan, M., Majhi, R. K., & DT, S. (2025).** “A Deep Learning Approach to Predict Surface Soil Wetness and Its Uncertainty Analysis Over the Tel River Basin, India.” *CLEAN–Soil, Air, Water, 53(1), e70003. (IF: 2.8).*
- **Sethi, R. R., Dandapat, A. K., Sankalp, S., Panda, D. K., Jena, S. K., & Sarangi, A. (2024).** “Analyzing time series variations in groundwater levels within the Baitarani River Basin, India.” *Environmental Earth Sciences, 83(12), 363. (IF: 2.8).*
- **Sahoo, B. B., Sankalp, S., & Kisi, Ozgur. (2023).** "A novel smoothing-based deep learning time-series approach for daily suspended sediment load prediction". *Water Resources Management. (IF:4.426).*

- **Sankalp, S., & Sahoo, S. N. (2023).** "An Automated Approach to Establish Relationship between Total and Effective Impervious Areas for Urban Indian Catchments." *Urban Water Journal* (IF: 2.675).
- **Sethi, R. R., Dandapat, A. K., Sankalp, S., Jena, S. K., Panda, D. K., & Sahoo, D. C. (2023).** "Innovative trend analysis for evaluation of groundwater storage in Baitarani River Basin." *Environmental Earth Sciences*, 82(24), 605. (IF: 2.8).
- **Sankalp, S., & Sahoo, S. N. (2022).** "Fuzzy AHP modelling of urbanization and environmental stress to rank selected Indian cities for liveability." *Environment, Development, and Sustainability*, 1-24. (IF: 4.080).
- **Sankalp, S., Sahoo, B. B., & Sahoo, S. N. (2022).** "Uncertainty and sensitivity analysis of deep learning models for diurnal temperature range (DTR) forecasting over five Indian cities." *Environmental Monitoring and Assessment* (IF: 3.307).
- **Sankalp, S., Sahoo, B. B., & Sahoo, S. N. (2022).** "Deep learning models comparable assessment and uncertainty analysis for diurnal temperature range (DTR) predictions over Indian urban cities." *Results in Engineering*, 13, 100326. (ESCI) (IF-5)
- **Arpan Pradhan, Kishanjit K. Khatua, and Sovan Sankalp (2018).** "Variation of Velocity Distribution in Rough Meandering Channels," *Advances in Civil Engineering*. (IF: 1.104).

Scopus Indexed Journals

- **Sethi, R. R., Dandapat, A. K., SK, S., Sankalp, S., RH, A., Bal, J. S., & Jothimani, M. (2026).** Entropy-weighted water quality index and TOPSIS for irrigation suitability assessment in eastern Odisha, India. *Water Science*, 40(1), 37.
- **Dandapat, A. K., Panda, P. K., Sankalp, S., Jothimani, M. (2025).** Evaluating Drought Risks and Meteorological Hazards in the Mahanadi Basin: Impact of Precipitation Deficits and Climate Change on the Downstream Regions of Hirakud Dam. *Discover Applied Sciences Volume 7, article number 717.* (IF-2.8)
- **Dandapat, A. K., Panda, P. K., Sankalp, S., Jothimani, M. (2024).** Quantifying Soil Erosion in Drought-Impacted Central Odisha, India, through Geospatial Mapping with RUSLE. *Applied and Environmental Soil Science*. (IF-2.4)
- **Dandapat, A. K., Panda, P. K., Sankalp, S., Jothimani, M. (2024).** Integrated Assessment of Groundwater Quality for Sustainable Irrigation in Drought-Prone Central Districts of Odisha, India. *Discover Sustainability*. 5, 238 (2024). (IF-2.1)
- **Sethi, R. R., Dandapat, A. K., Sankalp, S., Panda, D. K., Jena, S. K., & Patra, P. K. (2024).** "Integrating GIS-AHP Technique for Change Detection Analysis and Flood Susceptibility Mapping over the Baitarani River Basin". *Journal of Agricultural Engineering (India)*. NAAS Score (2024): 5.85.
- **Sahoo, B. B., Panigrahi, B., Nanda, T., Tiwari, M. K., Sankalp, S. (2023).** "Multi-step ahead urban water demand forecasting using deep learning models". *SN Computer Science*.
- **Sankalp, S., & Sahoo, S. N. (2022).** "Grey Relational Modelling of Land Surface Temperature (LST) for Ranking Indian Urban Cities". *Environmental Processes*, (IF-4.4).
- **Sethi, P. K., Sankalp, S., & Sahoo, S. N. (2021).** "Quantifying the dynamics of urban growth modes in Bengaluru, India". *Proceedings of the Institution of Civil Engineers-Urban Design and Planning*.
- **Sankalp, S., & Sahoo, S. (2021).** "Impact of urbanization on effective impervious area of Ahmedabad city in India." *ISH Journal of Hydraulic Engineering*.

Book Chapters (Scopus Published)

- *Sankalp, S., Rao, U.M., Patra, K. C & Sahoo, S. N. (2022).* "Modelling Gated Recurrent Unit (GRU) neural network in forecasting surface soil wetness for drought districts of Odisha". **Developments in Environmental Science, Elsevier.**
- *Sankalp, S and Panda, K.P. (2022).* "A comparative evaluation of Machine Learning and ARIMA models for forecasting relative humidity over Odisha districts". **Developments in Environmental Science, Elsevier.**
- *Kumar, Aman., Sankalp, S., Remesan. R. (2022).* "Spatiotemporal rainfall variability and trend analysis over all the districts of West Bengal during 1980-2021". **Developments in Environmental Science, Elsevier.**

Conference Proceedings (Scopus Published)

- *Arpan Pradhan and Sovan Sankalp (2023).* "Mapping of Built-up Area and Change Detection in Bengaluru using Semi-Automatic Classification". Novel Research Trends in Civil Engineering. **AIP Conference Proceedings.**
- *A K Dandapat, S Sankalp, P K Panda (2023)* "A topographical investigation of twenty-nine geomorphological parameters over Kuladara watershed, Odisha". Novel Research Trends in Civil Engineering. **AIP Conference Proceedings.**
- *S S Ojha, M Bhushan, S Sankalp, B B Sahoo, A Pradhan (2023)* "Addressing challenges and opportunities in enhancing water quality for irrigation. **AIP Conference Proceedings.**

Teaching Experience

- Assistant Professor at Centurion University of Technology and Management, Odisha from 2021-23 (Mentored 16 Undergrads for different Remote Sensing & GIS Projects)
- Associate Professor at Gandhi Institute for Education and Technology, Bhubaneswar, Odisha, India (2024-2025)

Achievements

- Successfully organized the International Conference on Civil, Environmental, and Applied Sciences (ICCEAS 2025) as Convenor. (Scopus Proceedings)(E3S Web of Conferences)
- International GIS DAY Celebrations 2021 Winner, as announced by ESRI, India for organizing as Convenor at Centurion University of Technology and Management, Paralakhemundi, Odisha.
- MHRD Fellowship by the Government of India for pursuing M.Tech and Ph.D.

International & National Collaborators

- Dr. Ozgur Kisi (Guest Professor, University of Applied Sciences, Lübeck, Germany)
- Dr. Aqil Tariq (Post Doctoral Fellow, Mississippi State University, USA)
- Dr. M. M Jothimani (Asst. Professor, Arba Minch University, Ethiopia)
- Prof. Upaka Rathnayake (Professor, Atlantic Technological University, Sligo, Ireland)
- Dr. Pankaj Kumar Gupta (Ramanujan Fellow, IIT Delhi)
- Dr. Suraj Jena (Principal Project Officer, AHRC, IIT Bhubaneswar)
- Dr. Ranu Rani Sethi (Principal Scientist, ICAR-IIWM)
- Dr. Arpan Pradhan (Asst. Professor, Christ University)
- Dr. Anindita Nath (Lecturer, Bankim Sardar College)
- Dr. Prafulla Kumar Panda (Dean SoET, Centurion University)
- Dr. Asit Kumar Dandapat (Senior Research Fellow, ICAR-IIWM)
- Prof. Sudhindra Nath Panda (Retd Prof, IIT Kharapur and Senior Advisor, AHRC, IIT Bhubaneswar)